Inequalities

Problem 1. Find the greatest real number C such that

$$\sqrt{\frac{x}{y+z}}+\sqrt{\frac{y}{z+x}}+\sqrt{\frac{z}{x+y}}>C$$

for any positive real numbers x, y, z.

Problem 2. Let $n \geqslant 2$ be an integer and let $1 < a_1 \leqslant a_2 \leqslant \cdots \leqslant a_n$ be n real numbers such that $a_1 + a_2 + \cdots + a_n = 2n$. Prove that

$$a_1a_2\cdots a_{n-1} + a_1a_2\cdots a_{n-2} + \cdots + a_1a_2 + a_1 + 2 \leqslant a_1a_2\cdots a_n.$$

Problem 3. Let $A = (a_{jk})$ be a 10×10 array of positive real numbers such that the sum of numbers in each row as well as in each column is 1. Show that there exist $j < k$ and $\ell < m$ such that

$$a_{j\ell}a_{km} + a_{jm}a_{k\ell} \geqslant \frac{1}{50}.$$

Problem 4. Given a convex polygon with $n \geqslant 3$ sides, let $a_1, a_2, \ldots, a_n$ be its side lengths, and let p be its perimeter. Find the smallest constant C such that

$$\frac{a_1}{p-a_1}+\frac{a_2}{p-a_2}+\cdots+\frac{a_n}{p-a_n}<C.$$

Problem 5. Let $\{a_n\}$ be a sequence of real numbers such that $a_1 = 2$ and $a_{n+1} = \dfrac{2a_n^2 + a_n + 1}{a_n - 1}$ for any positive integer n. Prove that $\displaystyle\sum_{k=1}^{n} \frac{1}{a_k} < \frac{2}{3}$ for any positive integer n.

Problem 6. Considering all possible choices of the positive integer n and the positive real numbers $a_1, a_2, \ldots, a_n$ with sum 1, find the smallest real number C such that

$$\frac{a_1}{(1+a_1)^2}+\frac{a_2}{(1+a_1+a_2)^2}+\cdots+\frac{a_n}{(1+a_1+a_2+\cdots+a_n)^2}<C.$$

Problem 7. Let $0 < x < \pi$ and let n be a positive integer. Prove that

$$\frac{\sin x}{1}+\frac{\sin 2x}{2}+\cdots+\frac{\sin nx}{n}>0.$$

Problem 8. Prove that there exists a real number C satisfying the following property: for any integers $n \geqslant 2$ and $X \subseteq \{1, 2, \ldots, n\}$, if $|X| \geqslant 2$, then there exist $x, y, z, w \in X$ (which can be the same) such that

$$0 < |xy - zw| < C\alpha^{-3},$$

where $\alpha = \dfrac{|X|}{n}$.

(Note: Weaker results with $C\alpha^{-3}$ replaced by $C\alpha^{-\beta}$ may be awarded points depending on the value of the constant $\beta > 3$.) :P